

# Project: EDA of Titanic Dataset

The RMS Titanic was a British passenger liner that sank in the North Atlantic Ocean in the early morning hours of 15 April 1912, after it collided with an iceberg during its maiden voyage from Southampton to New York City.

## 1. Problem Definition:

How can we comprehend the connections between the many passenger attributes in the dataset (such as age, gender, class, etc.) and their likelihood of surviving the incident? Finding correlations within the dataset is the goal of this EDA.

## 2. Data description:

The dataset consist 1309 entries with four (4) different columns: Unnamed: 0, sex, age and passengerClass.

## 3. Approach:

- Examine the data for missing values
- Analyze the relationship between features

## Import necessary libraries

```
In [1]: import numpy as np # for numerical computation
import pandas as pd # for data manipulation
import matplotlib.pyplot as plt # for data visualization
import seaborn as sns

''' magic command that ensures matplotlib plots
are displayed directly inside the Jupyter Notebook cells '''
%matplotlib inline
```

```
In [2]: # define a variable to store the dataframe
data = pd.read_csv('Titanic.csv')
```

```
In [3]: # print the dataframe to the console
data
```

Out[3]:

|      | Unnamed: 0                      | survived | sex    | age     | passengerClass |
|------|---------------------------------|----------|--------|---------|----------------|
| 0    | Allen, Miss. Elisabeth Walton   | yes      | female | 29.0000 | 1st            |
| 1    | Allison, Master. Hudson Trevor  | yes      | male   | 0.9167  | 1st            |
| 2    | Allison, Miss. Helen Loraine    | no       | female | 2.0000  | 1st            |
| 3    | Allison, Mr. Hudson Joshua Crei | no       | male   | 30.0000 | 1st            |
| 4    | Allison, Mrs. Hudson J C (Bessi | no       | female | 25.0000 | 1st            |
| ...  | ...                             | ...      | ...    | ...     | ...            |
| 1304 | Zabour, Miss. Hileni            | no       | female | 14.5000 | 3rd            |
| 1305 | Zabour, Miss. Thamine           | no       | female | NaN     | 3rd            |
| 1306 | Zakarian, Mr. Mapriededer       | no       | male   | 26.5000 | 3rd            |
| 1307 | Zakarian, Mr. Ortin             | no       | male   | 27.0000 | 3rd            |
| 1308 | Zimmerman, Mr. Leo              | no       | male   | 29.0000 | 3rd            |

1309 rows × 5 columns

## Data Cleaning

In [4]: *# rename the columns appropriately for better understanding*  
`data.columns = ['Name', 'Survived', 'Gender', 'Age', 'Class']`

In [5]: *# check to see if the columns have been renamed*  
`data.head()`

Out[5]:

|   | Name                            | Survived | Gender | Age     | Class |
|---|---------------------------------|----------|--------|---------|-------|
| 0 | Allen, Miss. Elisabeth Walton   | yes      | female | 29.0000 | 1st   |
| 1 | Allison, Master. Hudson Trevor  | yes      | male   | 0.9167  | 1st   |
| 2 | Allison, Miss. Helen Loraine    | no       | female | 2.0000  | 1st   |
| 3 | Allison, Mr. Hudson Joshua Crei | no       | male   | 30.0000 | 1st   |
| 4 | Allison, Mrs. Hudson J C (Bessi | no       | female | 25.0000 | 1st   |

In [6]: *# check for the number of missing values*  
`data.isnull().sum()`

```
Out[6]: Name      0
        Survived  0
        Gender    0
        Age      263
        Class     0
        dtype: int64
```

```
In [7]: # drop the missing values
data.dropna()
```

```
Out[7]:
```

|      | Name                            | Survived | Gender | Age     | Class |
|------|---------------------------------|----------|--------|---------|-------|
| 0    | Allen, Miss. Elisabeth Walton   | yes      | female | 29.0000 | 1st   |
| 1    | Allison, Master. Hudson Trevor  | yes      | male   | 0.9167  | 1st   |
| 2    | Allison, Miss. Helen Loraine    | no       | female | 2.0000  | 1st   |
| 3    | Allison, Mr. Hudson Joshua Crei | no       | male   | 30.0000 | 1st   |
| 4    | Allison, Mrs. Hudson J C (Bessi | no       | female | 25.0000 | 1st   |
| ...  | ...                             | ...      | ...    | ...     | ...   |
| 1301 | Youseff, Mr. Gerious            | no       | male   | 45.5000 | 3rd   |
| 1304 | Zabour, Miss. Hileni            | no       | female | 14.5000 | 3rd   |
| 1306 | Zakarian, Mr. Mapriededer       | no       | male   | 26.5000 | 3rd   |
| 1307 | Zakarian, Mr. Ortin             | no       | male   | 27.0000 | 3rd   |
| 1308 | Zimmerman, Mr. Leo              | no       | male   | 29.0000 | 3rd   |

1046 rows × 5 columns

```
In [8]: # check for duplicated values
data.duplicated().sum()
```

```
Out[8]: 0
```

## Exploratory Data Analysis (EDA)

```
In [9]: # check the length of the dataset
len(data)
```

```
Out[9]: 1309
```

```
In [11]: # finding the average age
data['Age'].mean()
```

```
Out[11]: 29.881134512434034
```

```
In [12]: # finding the maximum age
data['Age'].max()
```

```
Out[12]: 80.0
```

```
In [24]: # finding the minimum age
data['Age'].min()
```

```
Out[24]: 0.166700006
```

```
In [13]: # returns the passenger with the max age information
data[data['Age'].max() == data['Age']]
```

```
Out[13]:
```

|    | Name                            | Survived | Gender | Age  | Class |
|----|---------------------------------|----------|--------|------|-------|
| 14 | Barkworth, Mr. Algernon Henry W | yes      | male   | 80.0 | 1st   |

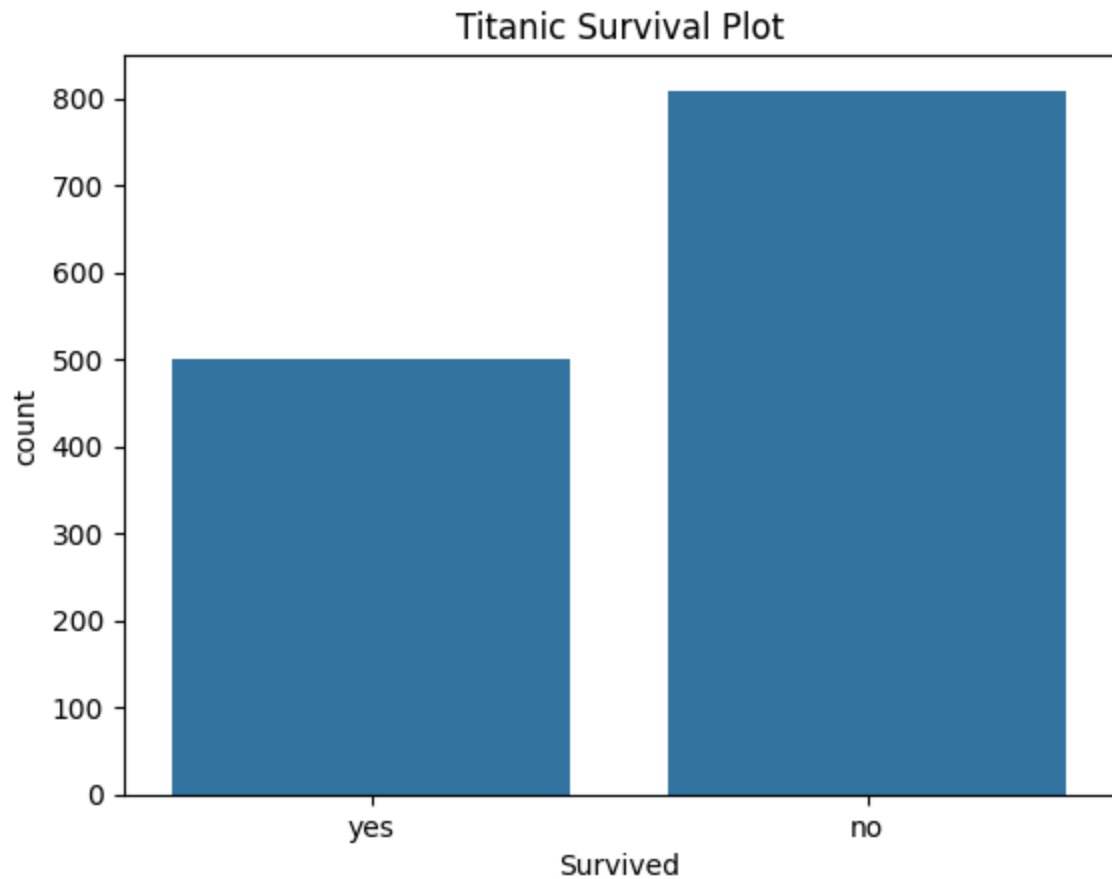
## Survived

```
In [14]: # number of passengers that survived and those who did not
data['Survived'].value_counts()
```

```
Out[14]: Survived
no      809
yes     500
Name: count, dtype: int64
```

```
In [15]: # -----
# creates a count plot showing the distribution of passengers
# who survived vs. did not survive.
# -----
sns.countplot(data, x = 'Survived')
plt.title('Titanic Survival Plot') # add title to the plot
plt.show # displays the countplot
```

```
Out[15]: <function matplotlib.pyplot.show(close=None, block=None)>
```

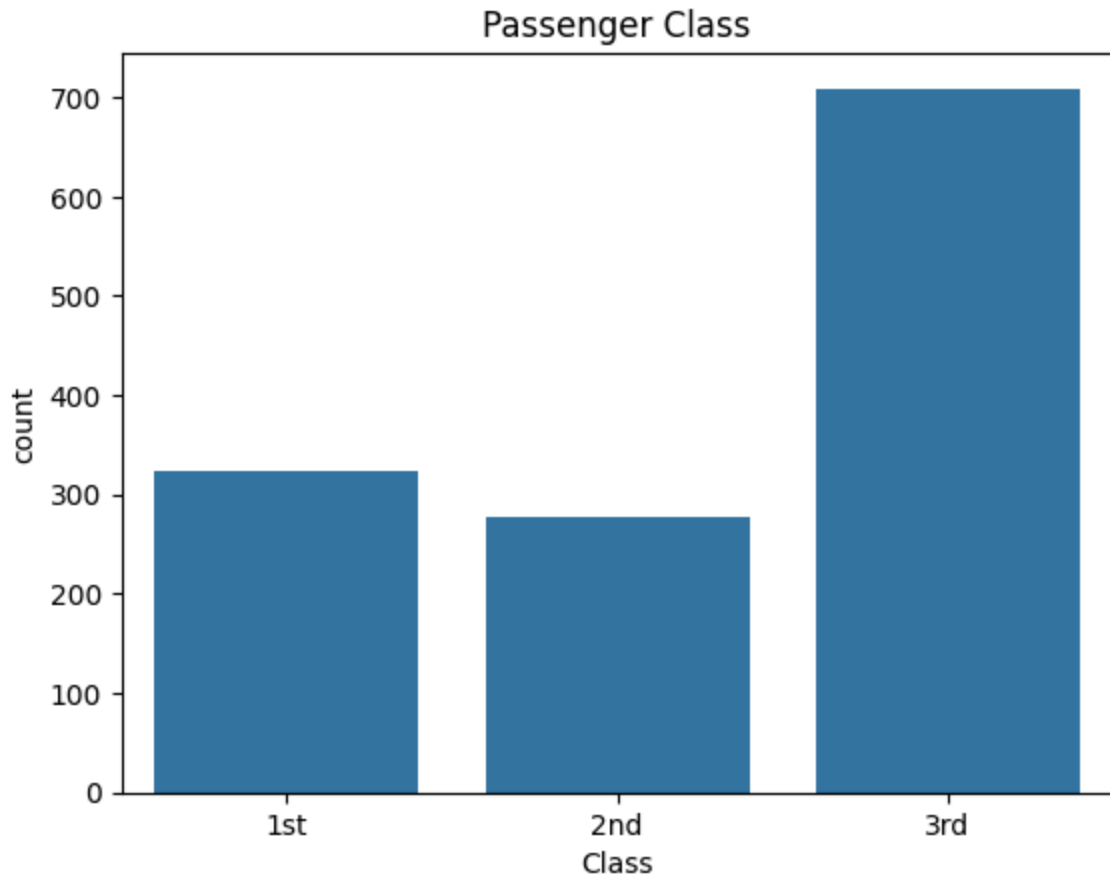


## Passenger Class

```
In [16]: # -----  
# returns the count of unique values in  
# the 'Class' column, sorted by frequency  
# -----  
data['Class'].value_counts()
```

```
Out[16]: Class  
3rd      709  
1st      323  
2nd      277  
Name: count, dtype: int64
```

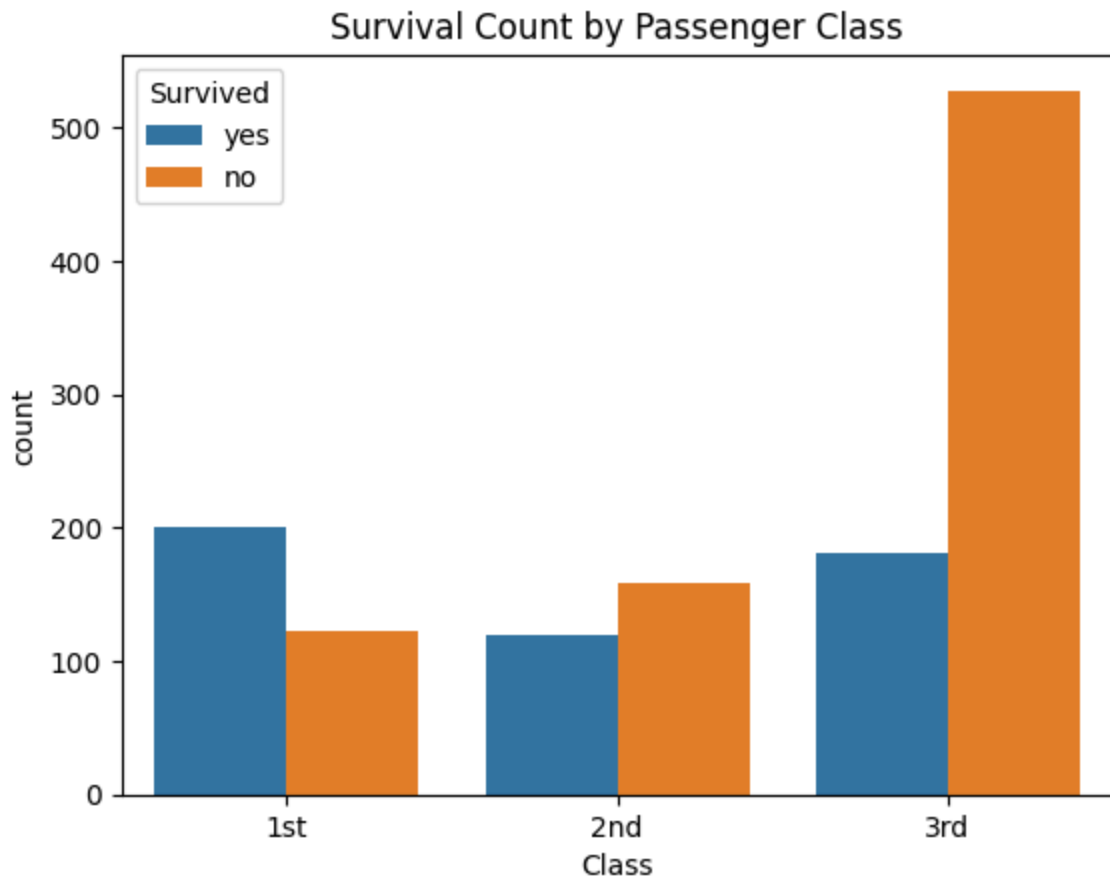
```
In [17]: # -----  
# creates a count plot showing the distribution of passenger class  
# -----  
sns.countplot(data, x = 'Class')  
plt.title('Passenger Class')  
plt.show()
```



```
In [18]: # -----  
# Groups the dataset by 'Class' and 'Survived',  
# then counts the number of occurrences in each group.  
# This shows how many passengers survived  
# or did not survive in each passenger class.  
# -----  
data.groupby(['Class', 'Survived'])['Survived'].count()
```

```
Out[18]: Class  Survived  
1st    no         123  
       yes         200  
2nd    no         158  
       yes         119  
3rd    no         528  
       yes         181  
Name: Survived, dtype: int64
```

```
In [19]: # -----  
# creates a count plot showing the number of passengers in each 'Class',  
# separated by survival status using the 'hue' parameter.  
# The plot helps visualize how survival rates differ across passenger classes.  
# -----  
sns.countplot(data, x = 'Class', hue = 'Survived')  
plt.title('Survival Count by Passenger Class') # adds a title to the plot  
plt.show() # display plot
```



## Gender

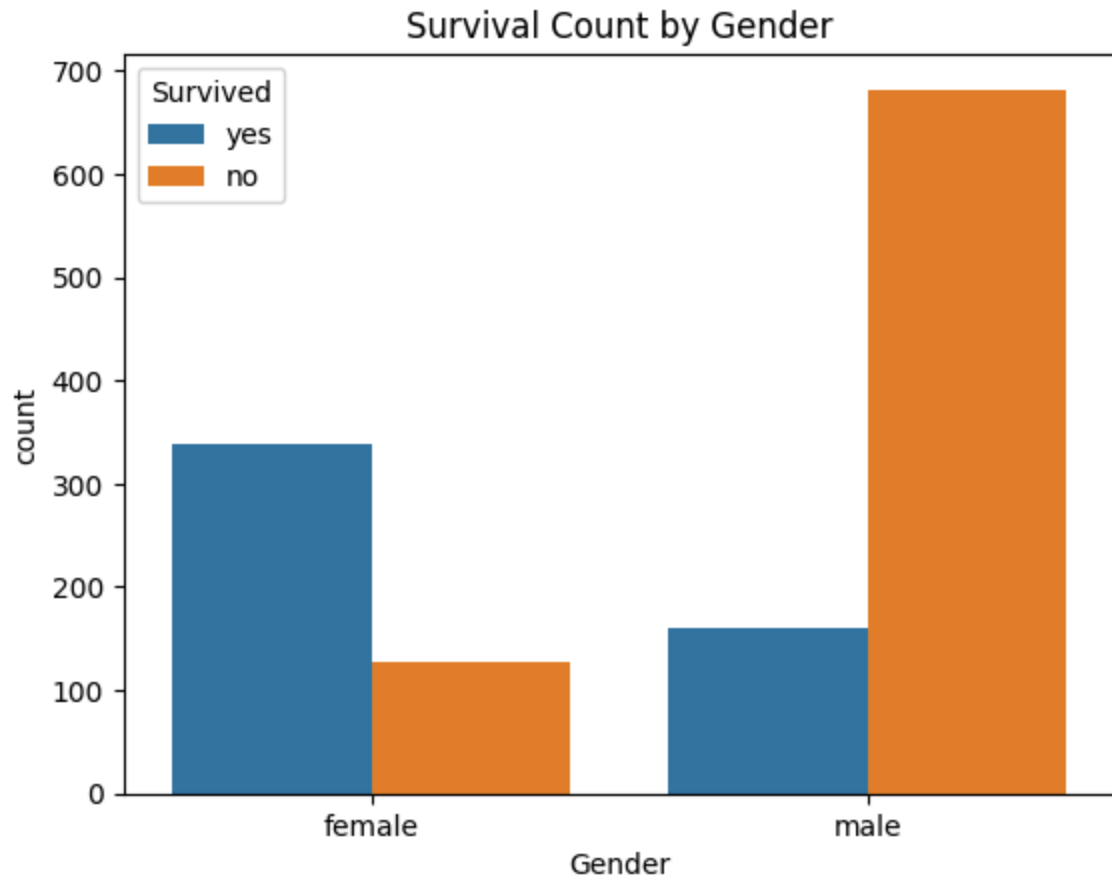
```
In [20]: # -----  
# returns the count of unique values in  
# the 'Gender' column, sorted by frequency  
# -----  
data['Gender'].value_counts()
```

```
Out[20]: Gender  
male      843  
female    466  
Name: count, dtype: int64
```

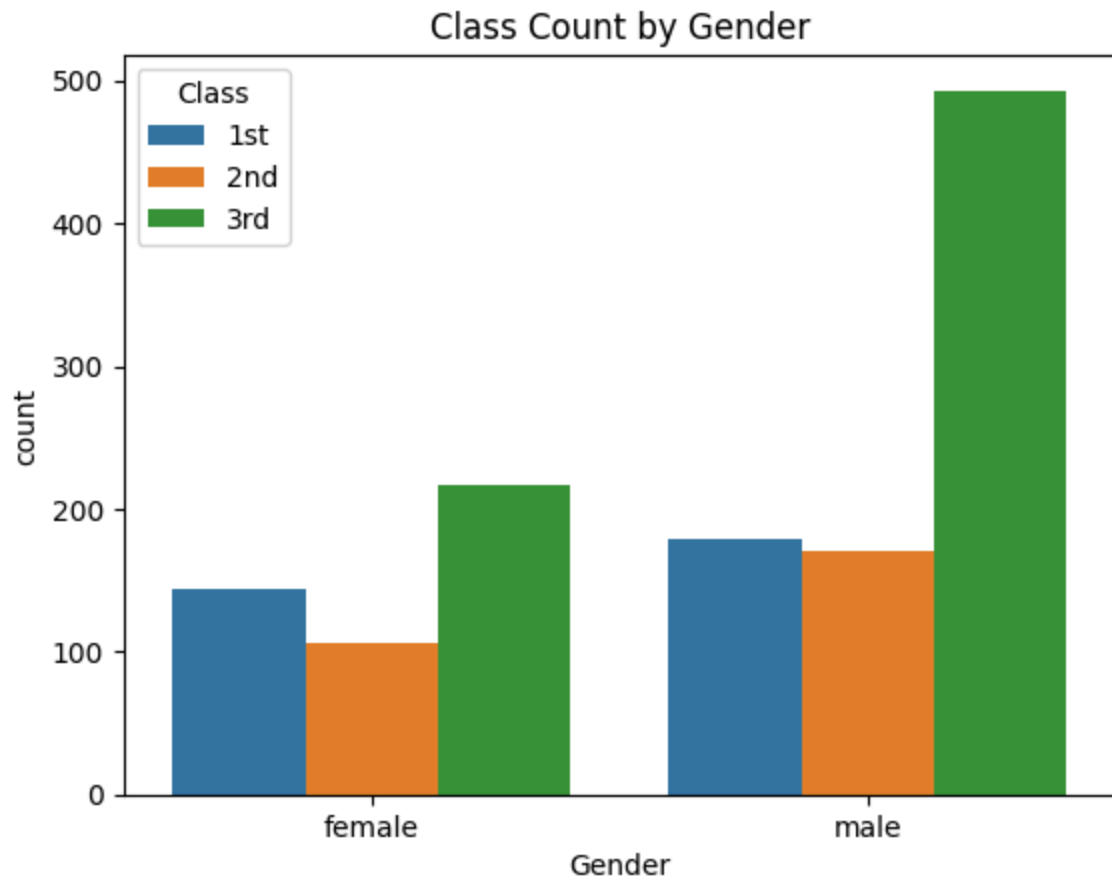
```
In [21]: # Groups the dataset by 'Gender' and 'Survived',  
# then counts the number of occurrences of each  
# survival status within each gender.  
data.groupby(['Gender', 'Survived'])['Survived'].value_counts()
```

```
Out[21]: Gender  Survived  
female  no         127  
         yes        339  
male    no         682  
         yes        161  
Name: count, dtype: int64
```

```
In [22]: # -----
# creates a count plot showing the number of passengers in each 'Gender',
# separated by survival status using the 'hue' parameter.
# The plot helps visualize how survival rates differ across passenger classes.
# -----
sns.countplot(data, x = 'Gender', hue = 'Survived')
plt.title('Survival Count by Gender') # adds a title to the plot
plt.show() # displays the plot
```



```
In [23]: # -----
# creates a count plot showing the number of passengers in each 'Gender',
# separated by passenger class using the 'hue' parameter.
# The plot helps visualize how survival rates differ across passenger classes.
# -----
sns.countplot(data, x = 'Gender', hue = 'Class')
plt.title('Class Count by Gender')
plt.show()
```



## Conclusion

In summary, the EDA suggests that **Gender**

and **Passenger Class** were the most significant factors determining survival.

**Female 1st Class passengers** would likely have the highest survival probability,

while **Male 3rd Class passengers** would likely have the lowest.

The overall fatality count was high, particularly among the

**largest passenger group (3rd Class)** and among **males**.

In [ ]: